GAUSE: Competition-Driven Specialization and Reward-Independent Capacity Assignment for Learner Populations

An Explanatory Companion: Architecture, Mechanisms, Demonstrations, and Applications

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Abstract

Learning systems that serve many environmental regimes with capacity-limited learners face a persistent allocation problem: which learner handles which regime, and—when the environment is non-stationary—who *keeps* a regime that falls dormant? We present **GAUSE** (Generalist-averse Affinity Update for Specialist Emergence), a deliberately minimal population-learning framework with three tightly coupled components: *per-regime belief learning* (Thompson sampling over a shared method inventory), *niche-affinity tracking* (a canonical exponentiated-gradient update on the regime simplex), and *winner-take-all competitive exclusion* (only the per-step winner reinforces its niche). At its core is one emergent property: the converged regime-to-agent assignment is *reward-independent*, so a specialist idles while its niche is dormant and retains its expertise. This property, not specialization per se, is what separates mechanisms that survive non-stationarity from those that do not. Across five capacity-allocation arms (capacity-bounded monolith, learned Mixture-of-Experts router, fixed random niches, an EOI/CDS-style diversity objective, and emergent competition), every reward-chasing allocation forgets dormant regimes while every reward-independent one retains the regimes it covers—and competition secures both at once (∼70% lower post-reactivation error than the reward-driven router, robust to swapping hard eviction for soft interference), recovering an *oracle fixed-assignment* skyline without being handed the assignment.

The comparison above is *task-incremental* (every arm sees the regime label). Dropping the label, a class-incremental variant that infers the regime from competence alone still retains on synthetic streams (0.38 post-reactivation vs. 1.07 for a label-free forgetting monolith). We then test it on *real* data (UCI Gas Sensor Array Drift) and report the outcome honestly: the label-free recovery *collapses* under real feature overlap, bounding the class-incremental claim to *separable-signal* regimes; and a full-capacity online classifier does *not* catastrophically forget on these features—relocating the forgetting story to the representation-sharing *neural* regime. On real data, accordingly, the contribution is a *mechanism and lens*, not a performance win.

The remaining evidence supports that reading. Across six real-world domains (145,294 records) and synthetic tasks the framework self-organizes into one-per-regime specialists (mean SI = 0.65 from competition alone) and matches purpose-built diversity and routing baselines on coverage with none of their machinery. Four further experiments close the design space: the router's forgetting reappears with *neural experts* on two standard benchmarks (permuted-digits +62%, Split-CIFAR-100 +22%); a *hybrid* reservation router isolates reward-independence of *protection* as the operative property; and drift and population-sizing sweeps map the mechanism's failure modes. We close with an evidence-tiered survey of applications (regime-switching finance, edge sensing, and elastic memory for agentic-AI / mixture-of-experts systems). Code and data: https://github.com/HowardLiYH/GAUSE.

Keywords: emergent specialization; competitive exclusion; multi-agent learning; catastrophic forgetting; mixture-of-experts; reward-independent assignment

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Contents

1 Introduction

"Two species with identical ecological niches cannot stably coexist."
—the competitive exclusion principle [5, 8]

Populations of learners that share an environment—trading strategies in a market, sensors in a distributed network, experts inside a mixture-of-experts model—face a coordination problem that single learners do not: how should the population *divide* a heterogeneous task space, without a central coordinator, so that every regime of the environment is served by a competent specialist? Despite decades of progress in multi-agent learning [27], three intertwined limitations persist in standard approaches.

First, **shared objectives homogenize**. Cooperative multi-agent reinforcement learning methods (IQL [23], VDN, QMIX [22], MAPPO [26]) optimize a shared task return, and a well-documented consequence is that agents converge to near-identical policies. In our head-to-head re-runs, all four methods reach a Specialization Index below 0.02 on every domain—no division of labor emerges by default. Second, **explicit coordination is expensive**. Methods that do produce diverse roles—diversity-MARL (ROMA [24], RODE [25], MAVEN [18], EOI [9], CDS [14]), quality-diversity archives [19, 21], or learned gating networks—require auxiliary machinery: intrinsic identifiability rewards, behavior descriptors chosen *a priori*, or a trained router. Each adds objectives, hyperparameters, and failure modes of its own. Third, and least appreciated, **reward-driven capacity allocation forgets** [3, 11]. When the environment is non-stationary and per-learner capacity is bounded, any mechanism that allocates capacity by chasing current reward—a single capacity-bounded model, or a Mixture-of-Experts gate trained on task reward—systematically overwrites knowledge of regimes that are currently dormant, and pays a full relearning cost each time a regime reactivates. The dormant regime is precisely the one emitting no reward signal, so a reward-driven allocator receives no signal to protect it.

To address these limitations, we introduce a cohesive design philosophy centered on a single ecological mechanism: **competitive exclusion** [5, 8, 17]. The framework, named **GAUSE** (Generalist-averse Affinity Update for Specialist Emergence)[1], is built from three integrated components (Figure 1):

First, each agent maintains *per-regime method beliefs*—Beta distributions over the success of each prediction method in each regime, acted on by Thompson sampling. This is the agent's competence: what it knows how to do, and where. **Second**, each agent tracks a *niche affinity* distribution over regimes, updated by the canonical exponentiated-gradient (Hedge / multiplicative-weights) rule [12, 4, 1]. This is the agent's identity: where it belongs. **Third**, agents are coupled solely through *winner-take-all competition*: at each step, all agents act, and only the best performer reinforces its affinity for the current regime. No communication, no shared reward, no gate, no diversity objective. Competition makes homogeneous strategies unstable—identical agents split the same wins, so deviating to a less-contested regime pays—and the population partitions into one-per-regime specialists.

The key insight of this document is that the resulting assignment has a property that purpose-built alternatives lack: *it is reward-independent at convergence.* A converged specialist holds its niche by identity, not by moment-to-moment reward, so when its regime goes dormant the specialist simply idles and *retains* its expertise (Figure 4). We formalize this as a **reward-independence principle** (Section 7), show that it cleanly dichotomizes five capacity-allocation mechanisms in controlled experiments (every reward-chasing arm forgets; every reward-independent arm retains), and demonstrate that it independently corroborates recent mixture-of-experts theory [15], which finds that the gating network's updates must be *terminated* for continual-learning convergence.

In summary, this work makes the following key contributions:

1) Catastrophic forgetting reframed as a signal-availability problem. Our most portable contribution is a lens, not an algorithm. Under bounded capacity, the quantity that most needs protecting—a dormant regime's knowledge—is exactly the one emitting no reward, so the protective signal is structurally correlated with the activity signal. Any allocator that protects capacity using current reward is therefore *blind* to dormant regimes by construction. We turn this observation into a scoped **reward-independence principle** (Principle 1): within the class of mechanisms whose only protection is the assignment itself, reward-independence is *necessary* for retention, and reward-independence *plus persistent coverage* is *sufficient*. We

[1] The name also honors G. F. Gause, whose competitive exclusion principle (1934) the mechanism operationalizes.

verify the prediction across five allocation mechanisms, confirm it transfers to *gradient-trained neural experts* on two standard continual-learning benchmarks (permuted-digits and Split-CIFAR-100; the failure is architecture-agnostic and benchmark-portable), bound it with a real-data reality check (Section 8.10: on near-separable real features a full-capacity learner does *not* forget, localizing the forgetting claim to the representation-sharing regime), and stress-test it with a *hybrid* reward-driven router carrying an explicit reservation term, which retains exactly when its protection becomes reward-independent. The GAUSE algorithm is then one instantiation that reaches a reward-independent, fully-covering assignment *for free*, where prior art needs a hand-tuned freeze schedule or an auxiliary diversity reward.

2) A coordination-free specialization mechanism. GAUSE couples winner-take-all competition with an exponentiated-gradient niche update and nothing else. Competition alone (niche bonus $\lambda = 0$) drives mean SI to 0.65 across six real domains; with the default bonus the population reaches SI ≈ 0.99 and one-per-regime coverage.

3) Honest benchmarking against purpose-built baselines. Rather than compare only to cooperative MARL (which does not try to specialize), we compare to an EOI/CDS-style learned-diversity objective, a learned MoE router, and a hybrid router with a memory-preservation term, all at matched capacity. Spatial coverage proves to be a commodity all arms achieve; competition's value there is parsimony. Retention is not a commodity: the reward-driven router fails it decisively, in both tabular and gradient-trained form.

4) A meaningfulness audit. We show that the Specialization Index certifies *organization*, not exploitable structure (it is invariant to regime-label shuffling and inflatable by the learning rate), and we localize exactly when division of labor improves task accuracy—under bounded capacity and non-stationarity—and when it does not. The audit extends to the mechanism's own failure modes: under *intra-regime drift* the converged assignment pins stale specialists (worse than a relearning monolith) until a staleness trigger re-opens competition, and scarce populations ($N < R$) under-cover regardless of per-agent capacity.

5) Matching the oracle ceiling without the assignment. Since every arm is handed the regime label, the honest competitor is an *oracle fixed assignment* given both the labels and a perfect covering permutation—the retention skyline. We show GAUSE recovers it without being handed the assignment: within 0.030 of the floor at K=1 and statistically indistinguishable for $K \geq 3$, and indistinguishable at *every* K under the soft memory model. Competition's value is automatic, pre-commitment-free recovery of the oracle partition, not beating it.

6) The retention result survives dropping the label (class-incremental). All of the above is task-incremental (the regime label is given). We then remove it: a variant that infers the active regime from competence alone still self-organizes against the *hidden* regimes (SI 0.75, coverage 0.86) and retains (0.38 post-reactivation vs. 1.07 for a label-free forgetting monolith), detecting reactivation purely from input similarity. On *real*, feature-rich data with genuine overlap (UCI gas sensor; Section 8.10), however, this label-free recovery *collapses* (post-reactivation $\to 0$, and the dissociation against the label-free monolith vanishes). We therefore report the class-incremental result honestly as a proof of concept bounded to *separable-signal* regimes—demonstrated where a separating signal exists, not a general capability.

2 Related Work

We review prior work along five dimensions closely related to GAUSE.

Cooperative MARL. Independent Q-Learning [23], value-decomposition methods (VDN, QMIX [22]), and multi-agent policy gradients (MAPPO [26]) optimize shared objectives with centralized training. These methods are the right baselines for cooperative control, but they do not attempt diversity, and empirically they homogenize: in our re-runs all reach SI $\leq$ 0.02. We treat them as confirmatory contrast, not as competitors on specialization.

Diversity- and role-emergence MARL. ROMA [24] and RODE [25] learn role embeddings; MAVEN [18] injects diverse exploration; EOI [9] and CDS [14] add intrinsic individuality or mutual-information objectives; R3DM [6] discovers roles through dynamics models. These methods do produce division of labor, at the cost of explicit diversity machinery. We benchmark directly against an EOI/CDS-style intrinsic-reward arm and find it a strong baseline: it matches competition on coverage for $K \geq 2$ and retains dormant regimes (its assignment is driven by an identity reward, not task reward—the reward-independence principle again). Competition's advantage over it is parsimony, not dominance.

Quality-diversity optimization. MAP-Elites [19] and novelty search [13] maintain archives of diverse high performers, but require behavior descriptors defined *a priori*. GAUSE needs no descriptors: niches are discovered by the dynamics.

Mixture-of-experts in continual learning. Recent theory [15] establishes that MoE can mitigate catastrophic forgetting [3, 20] by isolating tasks in experts—*provided* experts are plentiful relative to tasks and the gating network's updates are eventually terminated, without which continual task arrival destabilizes routing. Our experiments occupy exactly the regime those provisos exclude (fixed total capacity, experts forced into reuse, gate still learning) and show that isolation then breaks down: the router inherits the monolith's forgetting. Freezing the gate is, in our terms, making the assignment reward-independent; competition reaches that state emergently, with no freezing schedule to tune.

Ecological niche theory. The competitive exclusion principle [5, 8] and MacArthur's resource partitioning [17] describe how competition alone drives species into complementary niches [7]. GAUSE operationalizes this for artificial learners and inherits its most useful property: the partition, once formed, is maintained by identity rather than by ongoing payoff, which is what makes it robust to dormancy.

3 The GAUSE Framework: A Unified Architecture for Emergent Division of Labor

This section presents the integrated architecture. We first articulate the design philosophy (3.1), then detail the architectural blueprint (3.2), and finally explain the per-step dynamics that realize self-organization (3.3). Figure 1 contrasts the standard reward-driven allocation stack with the GAUSE architecture.

3.1 Design Philosophy: The Competitive-Exclusion Loop

Contemporary approaches to division of labor treat the assignment of learners to tasks as something to be *optimized*: a gate is trained, a diversity objective is added, roles are learned. GAUSE is founded on the opposite principle: **the assignment should be an equilibrium, not an objective**. We formalize this as the competitive-exclusion loop, a recursive paradigm structured around four functions (Figure 2):

Compete: all agents act on the current regime; one wins.
Reinforce: only the winner strengthens its affinity for that regime.
Differentiate: losers' relative affinities drift toward less-contested regimes.
Stabilize: once each agent owns a regime, deviation is unprofitable; the partition is a fixed point.

The key property of this loop is *what it leaves behind*. Because the converged partition is a fixed point of identity (each agent's affinity concentrates on its niche), it no longer depends on the flow of reward. An agent whose niche is dormant receives no wins, but it also receives no pressure to abandon its niche—it simply idles. This is the structural source of the retention results in Section 8: the loop intrinsically unifies *organizing* with *remembering*.

3.2 Architectural Blueprint: Components and Their Roles

The loop is instantiated through three principal components per agent, plus one population-level rule (Figure 1b).

1. Method-Belief Store: The Competence Layer. Each agent i maintains, for each regime r it has encountered, a Beta distribution $\mathrm{Beta}(\beta^+_{i,r,m}, \beta^-_{i,r,m})$ over the success rate of each method m in a shared inventory of M methods. This store answers *"what works here?"* and is updated by standard Bayesian counting. In the bounded-capacity experiments the store is explicitly finite: it holds at most K regimes, with either least-recently-used eviction (hard model) or graded interference decay (soft model)—Section 6.

2. Niche Affinity: The Identity Layer. Each agent maintains a distribution $\alpha_i \in \Delta^R$ over regimes—its evolving answer to *"where do I belong?"* Affinity is updated only on wins, by the canonical exponentiated-gradient rule (Section 4), and read out through the Specialization Index $\mathrm{SI}(\alpha) = 1 - H(\alpha)/\log R$, which is 1 for a one-hot specialist and 0 for a uniform generalist.

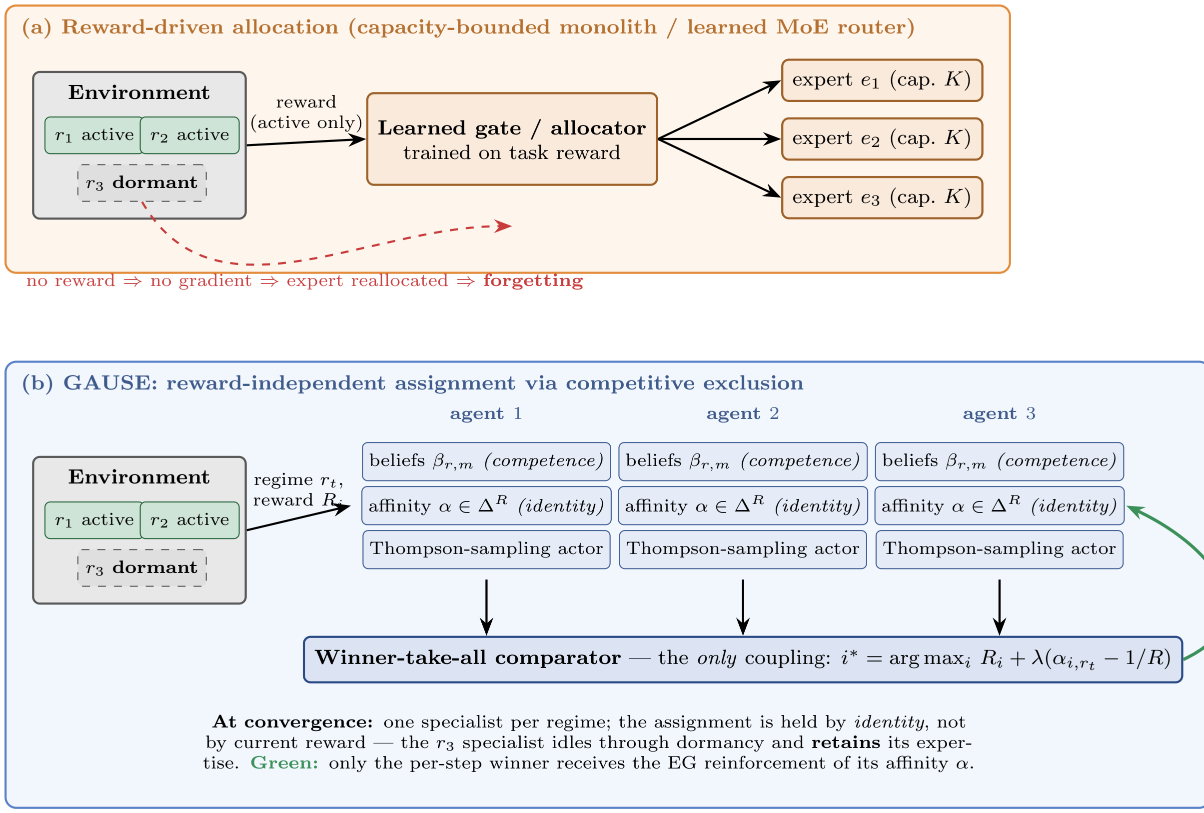

Figure 1 The GAUSE architecture, contrasted with reward-driven allocation. **(a)** A gate (or a single bounded model) assigns capacity by chasing current task reward; a dormant regime emits no reward gradient, so its capacity is silently reallocated and must be relearned on reactivation. **(b)** GAUSE couples agents only through per-step winner-take-all competition; each agent's niche affinity α concentrates on the regime it wins, and the converged one-per-regime partition is *structural*—it persists with no reward flow, which is the source of the retention results in Section 8.

3. Thompson-Sampling Actor: The Decision Layer. At each step the agent samples $\tilde{\theta}_m \sim \mathrm{Beta}(\beta^+, \beta^-)$ for every method and plays the argmax—a principled exploration–exploitation balance requiring no tuned exploration schedule.

4. Winner-Take-All Rule: The Population Coupling. The *only* interaction between agents is the per-step competition: the agent with the highest (optionally niche-shaped) reward is declared winner and is the only one to reinforce its affinity. There is no communication channel, no shared critic, no gate, and no diversity term. This minimality is deliberate: every emergent property of the system must be attributable to competition, because nothing else couples the agents.

3.3 System Dynamics: One Step of the Loop

The components interact in a fast per-step cycle. Given the environment in regime r_t:

1. **Act.** Every agent i selects a method m_i by Thompson sampling from its beliefs for r_t (agents without competence in r_t act uninformed) and receives raw reward R_i.
2. **Score.** Each agent's competitive score adds an additive niche bonus centred on the uniform affinity: $\tilde{R}_i = R_i + \lambda\,(\alpha_{i,r_t} - 1/R)$, with $\lambda \geq 0$. At $\lambda = 0$ the bonus vanishes and raw reward decides.
3. **Select winner.** $i^* = \arg\max_i \tilde{R}_i$.
4. **Update beliefs.** Agents update their Beta counts for the methods they played (success/failure), within their capacity budget.

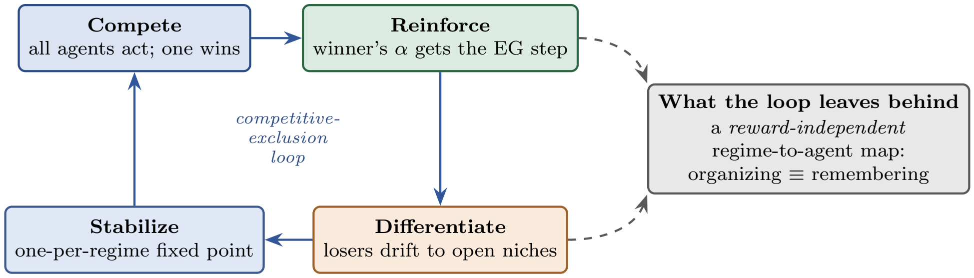

Figure 2 The competitive-exclusion loop. Compete–Reinforce–Differentiate iterates until Stabilize: each agent owns a regime and deviation is unprofitable. The loop's by-product is its most valuable output—a converged assignment that no longer depends on the flow of reward.

5. **Update identity.** Only the winner applies the exponentiated-gradient affinity update for r_t and renormalizes.

Two timescales emerge from this single cycle without being designed in: a *fast* competence dynamic (beliefs sharpen within a regime in tens of steps) and a *slow* identity dynamic (affinities concentrate over hundreds of steps). The slow dynamic is the one with the architectural consequence: once identities have concentrated, the regime-to-agent map is fixed and reward-independent, and the population behaves as a distributed, persistent memory over the regime space.

4 Niche Affinity and the Exponentiated-Gradient Update

4.1 The Update Rule

When agent i wins in regime r with normalized reward (quality) $q \in [0, 1]$, its affinity updates multiplicatively and renormalizes:

$$\alpha_{i,r} \leftarrow \frac{\alpha_{i,r}\, e^{\eta q}}{\sum_{r'} \alpha_{i,r'}\, e^{\eta q\, \mathbf{1}[r'=r]}}, \tag{1}$$

the canonical exponentiated-gradient / Hedge / multiplicative-weights step on the regime simplex [12, 4, 1, 2]. Three properties motivate this choice over additive heuristics. **(i) Simplex preservation by construction:** affinities remain a probability distribution with no clamping (an earlier additive variant of this project required an undocumented clamp whose pathology suppressed roughly half the specialization signal; replacing it with EG is what separates the current V4 results from the V3 ones). **(ii) A constant effective rate:** the multiplicative step does not slow down or speed up depending on the current state in the way the clamped additive rule did. **(iii) A standard form:** EG on the simplex is the textbook update for tracking the best coordinate, making the algorithm's one learnable dynamic maximally legible. We deliberately do *not* claim Hedge's adversarial regret bound transfers to our coupled winner-take-all setting; that would require separate analysis.

4.2 Worked Demonstration

Consider $R = 3$ regimes, $\eta = 0.6$, and an agent at uniform affinity $\alpha = (0.333, 0.333, 0.333)$. The agent wins three consecutive steps in regime 1 with quality $q = 1$:

$$(0.333, 0.333, 0.333) \;\rightarrow\; (0.477, 0.262, 0.262) \;\rightarrow\; (0.624, 0.188, 0.188) \;\rightarrow\; (0.751, 0.124, 0.124).$$

Three wins move SI from 0 to 0.33; ten wins take it past 0.95 (Figure 3). The same trajectory in reverse never occurs spontaneously: losing in regime 1 does not push the agent away (losers do not update); it is *winning elsewhere* that re-anchors identity. This asymmetry—identity moves only on positive evidence of fit—is what makes converged identities stable through dormancy.

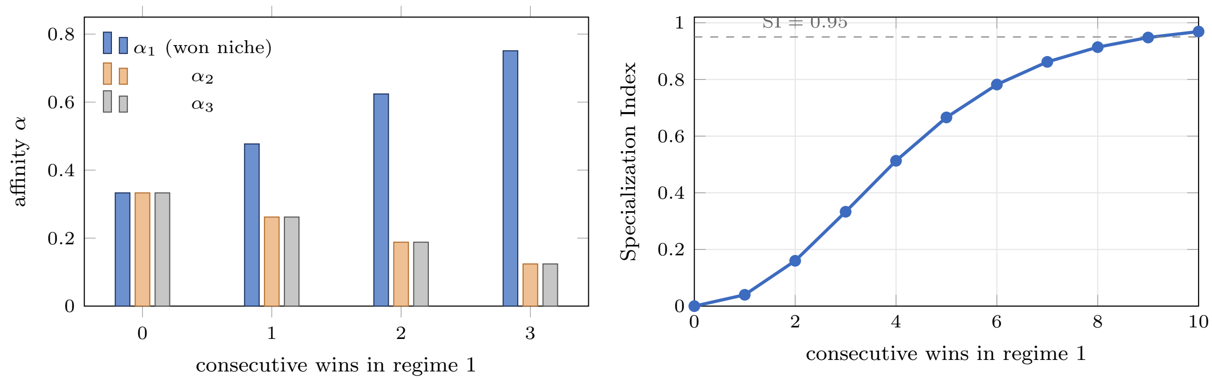

Figure 3 Worked demonstration of the exponentiated-gradient affinity update ($R = 3$, $\eta = 0.6$, quality $q = 1$). **Left:** the winning regime's affinity grows multiplicatively while the others shrink by renormalization. **Right:** the Specialization Index $\mathrm{SI} = 1 - H(\alpha)/\log R$ traced over ten consecutive wins—identity concentrates quickly, and only positive evidence (winning) ever moves it.

4.3 Heterogeneous Initialization and Symmetry Breaking

Agents start near-uniform with small random perturbations. The perturbation carries no labels and no assignment; it only breaks ties so that the first few competitions are not decided by coin flips alone. The eventual one-per-regime partition is an equilibrium selected by the dynamics, not an initialization artifact: random-shuffle controls confirm the same partition statistics under re-drawn perturbations.

5 Competitive Exclusion and Winner-Take-All Dynamics

5.1 Why Strictness Matters

Winner-take-all is not a simplification of softer competition—it is the key structural driver of differentiation. Under proportional (softmax) credit, identical agents can persist indefinitely: each receives a share of every regime's reward, and hedging across niches is stable. Under winner-take-all, identical agents split wins evenly in expectation, so each one's per-regime reinforcement halves; an agent that deviates to an uncontested regime wins that regime *outright*. Formally (companion paper [16], Proposition 1), homogeneous strategy profiles are not Nash equilibria of the winner-take-all game: a unilateral deviation to a less-contested niche strictly increases expected reinforcement.

5.2 The Niche Bonus Is an Accelerator, Not the Cause

The additive shaping term $\lambda(\alpha_{i,r_t} - 1/R)$ advantages an agent competing in a regime it already prefers. Setting $\lambda = 0$ removes it entirely—and specialization still emerges, with mean SI 0.65 across the six real domains (every domain > 0.49, versus 0.13 for a random baseline). The λ-sweep is flat and high ($\mathrm{SI} > 0.98$) across $\lambda \in [0.2, 0.4]$ and declines at 0.5, where over-locked winners stop contesting secondary regimes and leave some niches under-filled. The mechanism therefore needs no shaping to work; shaping only sharpens it.

5.3 What the Population Looks Like After Convergence

On every real domain, the converged population is a clean partition: each agent holds a primary niche with affinity ≥ 0.95, every regime is owned by at least one agent, and the population collectively uses 86% of the shared method inventory (specialists adopt the methods suited to their niche). The assignments are interpretable by inspection—a property no deep MARL baseline in our comparison offers.

6 Bounded Capacity and Memory Models

Specialization only matters when no single learner can do everything. We model this with an explicit per-agent capacity K: each agent can hold beliefs for at most K of the R regimes. Two memory models implement the constraint.

6.1 Hard Model: LRU Slots

The belief store holds K regime entries; learning an uncached regime evicts the least-recently-used entry. Eviction is total: the evicted regime's beliefs reset to the uninformed prior, and re-encountering it costs a full relearning period. This is the cleanest possible model of "capacity binds."

6.2 Soft Model: Interference Decay

A natural objection is that real learners do not evict discretely—knowledge *interferes* [3]. The soft model removes eviction entirely: every learning update on the active regime decays the pseudo-counts of the agent's other held regimes by a factor that grows with how far the agent's learned footprint exceeds K. Dormant regimes are never refreshed, so they decay smoothly toward the prior—the interference account of catastrophic forgetting, with graded rather than all-or-nothing loss. Every retention result in Section 8 is reproduced under both models; the headline separation is a property of *who gets updated*, not of how forgetting is discretized.

6.3 The Five Capacity-Allocation Arms

All experiments compare five mechanisms at matched per-agent capacity K, together with one privileged *oracle skyline* (arm 6) that we use as an upper reference rather than as a competing mechanism:

1. **Monolith**: one agent with capacity K serves everything. Allocation chases the active regime by necessity.
2. **MoE router**: N capacity-K experts behind a learned gate (ϵ-greedy routing; gate trained on routed-expert quality). Allocation chases current reward by design.
3. **Random diversity**: N agents with fixed, random capacity-K niches. Allocation is frozen—fully reward-independent, but coverage is accidental.
4. **Learned diversity (EOI/CDS-style)**: all agents receive an intrinsic identifiability reward $\log p(i \mid r)$ on top of task quality [9, 14]; the identified owner learns the regime. Allocation follows an *identity* signal, not task reward.
5. **GAUSE (ours)**: winner-take-all competition; allocation is an emergent, converged identity.
6. **Oracle fixed assignment (skyline)**: $N{=}R$ agents pinned to a hand-designed *covering* assignment—at $K{=}1$ the identity permutation (agent $i \mapsto$ regime i), with no learning of *where* to specialize. This arm is reward-independent and perfectly covering *by construction* and pays zero discovery cost, so it marks the best achievable retention; it is privileged because it is handed both the regime labels and the assignment.

This five-arm design is deliberate: the arms differ in *what signal drives capacity assignment*, which is exactly the axis the next section's principle predicts should matter. The oracle skyline adds the natural sixth question—if the labels are known, why not simply hand-assign?—which we answer directly in Section 7 and Table 2.

What each arm observes (and why the comparison is fair). All six arms receive the same information at each step: the current regime label r_t and the per-method quality signal. This is the *task-incremental* continual-learning setting (task identity known), not the harder *class-incremental* one (task identity must

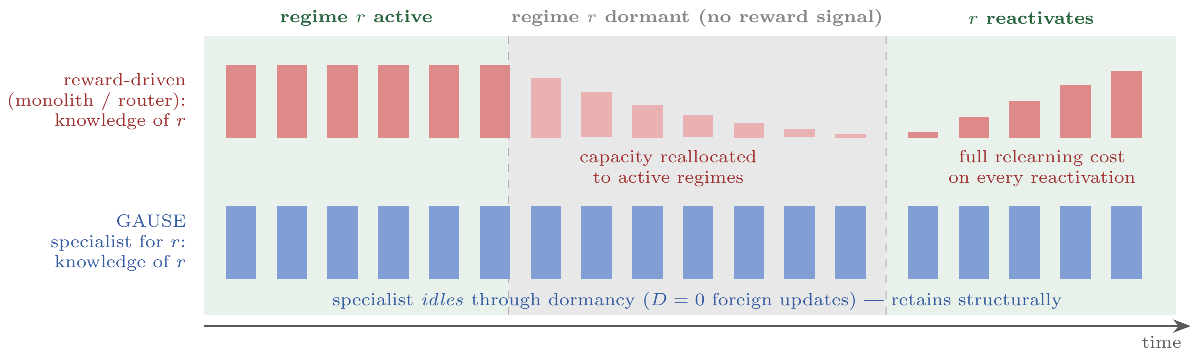

Figure 4 Why reward-independence governs retention. **Top:** a reward-driven allocator (capacity-bounded monolith or learned MoE router) reallocates the dormant regime's capacity to active regimes—the dormant regime emits no reward gradient, so nothing protects it—and pays a full relearning cost at reactivation. **Bottom:** a converged GAUSE specialist holds its niche by identity, applies no foreign updates to its own store during dormancy, and is immediately competent on reactivation. Measured post-reactivation errors for all five arms appear in Table 2 and Figure 6.

be inferred). We make this explicit because it bounds the claim and keeps the comparison information-symmetric: the dissociation below is *not* an artifact of GAUSE seeing the regime while the router must infer it—the MoE gate, the monolith, and every other arm are handed exactly the same r_t. What differs across arms is solely *what each does with that label*: whether the capacity assignment it induces chases current reward or is reward-independent. GAUSE's contribution is therefore scoped precisely: *given* task identity, a self-organized reward-independent assignment retains where reward-chasing ones forget. Removing the oracle label—inferring r_t from the competence beliefs themselves, which already form a per-regime likelihood—is the natural next step and we return to it in Section 10.

7 The Reward-Independence Principle

7.1 Why a Reward-Driven Router Forgets

Consider a regime r that goes dormant for an extended interval while an expert e holding r receives D learning updates from other, active regimes. Figure 4 illustrates the mechanism.

Observation 1 (Dormant regimes receive no protective signal)**.** *Under the hard capacity model, e retains r only if fewer than K distinct regimes are used during the interval; under the soft model, r's retained strength decays as $(1-\rho)^D \to 0$. Crucially, a dormant regime emits no steps and hence no reward gradient: the gate update $\Delta\,\mathrm{gate}[r][e] \propto (q_e - \frac{1}{2})$ is identically zero throughout dormancy, so the router receives no signal that would lead it to reserve e for r. Reactivation therefore incurs the full relearning cost with probability $\to 1$ as dormancy grows.*

The asymmetry is the crux: *routing and capacity protection draw on the same signal, and the regime most needing protection is the one producing none.* A converged competitive (or identity-driven) assignment escapes the trap because it is reward-independent at steady state: the specialist for r simply idles through dormancy, applying zero foreign updates to its own store ($D = 0$ for its niche), and retains it structurally—no signal required.

Principle 1 (Reward-independent assignment)**.** *Consider capacity-allocation mechanisms whose* only *means of protecting a dormant regime's knowledge is the regime-to-learner assignment itself—that is, mechanisms with no auxiliary memory-preservation, rehearsal, or capacity-reservation term (the monolith, the vanilla learned router, fixed/learned/competitive niches all lie in this class). Within this class, under bounded per-agent capacity and non-stationarity:*

(i) ***(Necessity)*** *reward-independence of the assignment is* necessary *for retention: any allocation that reassigns capacity as a function of current task reward reallocates a dormant regime's capacity with*

probability $\to 1$ as the dormancy interval grows, because that regime emits no reward (Observation 1); and

(ii) ***(Sufficiency)*** *reward-independence* together with persistent coverage—*every recurring regime keeps an assigned learner—is* sufficient *for retention: the assigned learner idles through dormancy and preserves its store.*

Scope of the principle (two honest caveats). *Reward-independence alone is not sufficient.* Persistent coverage is a separate requirement: *fixed random* niches are reward-independent yet can leave a recurring regime unowned at $K{=}1$, so they retain only the regimes they happen to cover (this is why random diversity retains only *partially* in Table 1 and Table 2). Competition's advantage among the reward-independent arms is precisely that winner-take-all exclusion *guarantees* coverage—it is the cheapest mechanism that satisfies both conjuncts at once. *Reward-dependence is not necessary for forgetting in general.* The necessity claim is scoped to the assignment-only class: a reward-driven router *augmented* with an explicit memory-preservation or capacity-reservation term lies outside the class and could in principle retain while still routing by reward. We test exactly this hybrid (Section 8.9, Figure 10) and confirm it: a router with a reservation term recovers most of the lost retention when spare capacity exists. Our results therefore identify *what extra objective a reward-driven router needs*—reward-*independent* protection of capacity—not an inherent limit of routing.

On the status of the "Principle." We are deliberate about register. Within the assignment-only class as *defined*, the sufficiency direction is close to analytic—"reward-independent and covering" is nearly the definition of "keeps a learner idling on each dormant regime." We therefore do not rest the contribution on the definition. Its force is twofold and empirical: (i) the necessity direction is a quantitative claim about a *rate*, which we make precise and prove below (Proposition 1) rather than assert, and (ii) real mechanisms that *could* behave otherwise line up with the prediction five-for-five (Table 1); and the hybrid router—which sits *outside* the five-arm class—independently corroborates it, partially retaining exactly when its protection becomes reward-independent. The principle is best read as an organizing observation whose content is carried by the dissociation and the rate bound, not by the taxonomy.

7.2 A Probabilistic Dormancy Model: the Necessity Claim Is Not Vacuous

The necessity direction asserts that a reward-driven allocator reallocates a dormant regime's capacity "with probability $\to 1$." We make this a theorem under an explicit model rather than leave it as intuition.

Model. Fix capacity $K < R$ and the hard (LRU) store. Let regime r go dormant at time 0. During dormancy the active regimes arrive as an i.i.d. stream; let U_t be the number of *distinct* regimes other than r that have been served and learned by the learner holding r up to time t. A reward-driven allocator (monolith, or MoE gate with ϵ-greedy routing) directs each active step to the learner that currently maximizes routed quality; because r emits no step, its gate logit receives the null update $\Delta\,\mathrm{gate}[r][e] \propto (q_e - \frac{1}{2}) = 0$ throughout dormancy (Observation 1), so the assignment of r is driven *only* by the competing active regimes. A reward-independent allocator never conditions the assignment on this stream at all.

Proposition 1 (Reallocation rate)**.** *Let $p > 0$ be the per-step probability that the learner holding r is made to learn a distinct not-yet-held active regime, and let T be the dormancy length.*

(i) ***(Reward-driven)*** *Under LRU eviction, r is evicted as soon as K distinct other regimes are learned, i.e. once $U_T \geq K$. Since U_T is non-decreasing with independent positive increments,* $\Pr[r \text{ retained}] = \Pr[U_T < K] \leq K\,(1-p)^{T-K+1}\binom{T}{K-1} \to 0$ *as $T \to \infty$ (a sum of K binomial terms, each bounded by $\binom{T}{K-1}(1-p)^{T-K+1}$); under the soft model the retained strength decays as $(1-\rho)^D \to 0$, where D is the number of foreign updates applied to r's store during dormancy (Observation 1; $D \geq U_T$, since each distinct regime contributes at least one update) and ρ the per-update interference rate. In both cases* $\Pr[\text{reallocation}] \to 1$.

(ii) ***(Reward-independent)*** *If the assignment of r does not depend on the active stream—the learner owning r applies $D = 0$ foreign updates to its own store while r is dormant—then* $\Pr[\text{reallocation}] = 0$ *for all T, and the store for r is preserved exactly.*

Proof sketch. (i) The number of dormancy steps before the K-th distinct foreign regime is learned is a sum of K geometric(p) waiting times (a negative-binomial), which is a.s. finite; hence $\Pr[U_T \geq K] \to 1$ and the displayed tail bound follows from the negative-binomial CDF. LRU evicts r on that event; the soft-model statement is immediate from the multiplicative decay applied once per foreign update. (ii) By hypothesis no foreign update touches r's store, so its pseudo-counts—and therefore the learner's competence on r—are unchanged at reactivation. □

The two cases are genuinely different functions of T (one $\to 1$, the other $\equiv 0$), so the necessity claim is not true by relabeling: it is a statement that any allocator whose assignment is a function of the active (reward-bearing) stream inherits the eviction rate of part (i), while only an allocator that makes the assignment a function of *identity* escapes to part (ii). GAUSE reaches the part-(ii) condition at convergence without being told to freeze.
Model vs. experiment. The bound assumes an i.i.d. stream of active regimes, whereas our experiments use a *deterministic* sliding window of W regimes per epoch. The idealization is conservative: under the deterministic schedule a dormant regime is guaranteed to see W active competitors every epoch, so U_T grows at least as fast as under the i.i.d. model and the eviction event $\{U_T \geq K\}$ arrives no later—the i.i.d. analysis therefore upper-bounds the true retention probability rather than flattering it.

7.3 The Five-Arm Dissociation

Principle 1 makes a sharp prediction across the five arms of Section 6, verified in full by the experiments (Table 1):

Arm	Assignment signal	Reward-indep.?	Covers?	Retains?
Monolith	current active regime	no	—	no
MoE router	current task reward	no	—	no
Random diversity	none (frozen)	yes	not guaranteed	partial
Learned diversity	intrinsic identity reward	yes	yes	yes
GAUSE	converged identity	yes	yes (guaranteed)	yes
Oracle fixed (skyline)	hand-assigned (given labels)	yes	yes (by design)	yes

Table 1 The five-arm dissociation predicted by Principle 1, plus the oracle fixed-assignment skyline (last row, privileged: given the labels *and* the assignment). Retention requires *both* conjuncts of the principle: reward-independence of the assignment *and* persistent coverage. The two reward-driven arms fail the first conjunct and forget. Among the reward-independent arms, fixed random niches satisfy reward-independence but not guaranteed coverage, so they retain only the regimes they happen to own (*partial*); the learned-diversity and competitive arms satisfy both and retain in full. Retention thus tracks the conjunction—not specialization, population size, or learning sophistication.

The pattern matches the principle exactly: every arm whose assignment chases current reward (monolith, router) forgets, and every reward-independent arm retains the regimes it covers. Coverage—the second conjunct—is what separates the reward-independent arms from one another. Among them, competition is the most parsimonious: random diversity is reward-independent but leaves coverage to chance (unowned regimes are possible at $K = 1$, hence only partial retention); learned diversity covers and retains but requires the auxiliary intrinsic-reward machinery; gate-freezing (the prescription of recent MoE continual-learning theory [15]) retains but requires deciding *when* to freeze. Competition needs none of these—winner-take-all exclusion is the equilibrium that delivers reward-independence *and* guaranteed coverage at once.

7.4 Relation to MoE Continual-Learning Theory

Recent theory on mixture-of-experts in continual learning [15] proves that the gating network's updates must be *terminated* after sufficient training for the system to converge, and that MoE's forgetting-mitigation holds when experts are plentiful or can be added. Read through Principle 1, both findings are the same statement: a stabilized (frozen) gate is a reward-independent assignment, and plentiful experts remove the forced reuse that makes reward-chasing destructive. Our experiments cover the complementary regime the theory excludes—fixed total capacity, forced reuse ($K \ll R$), gate still learning—and find exactly the failure the principle predicts. The two lines of work corroborate each other from opposite directions: regression theory prescribes freezing; population dynamics exhibits a mechanism that arrives at the frozen state emergently.

8 Experiments and Evaluation

8.1 Experimental Settings

Domains. Six real-world domains with verified public data (145,294 records): cryptocurrency (Bybit; 8,766 daily bars), commodities (FRED; 5,630), weather (Open-Meteo; 9,105), solar irradiance (Open-Meteo; 116,834 hourly), urban traffic (NYC TLC; 2,879 hourly), and air quality (Open-Meteo; 2,880 hourly). Each domain defines 4–6 regimes and 5 tailored prediction methods. Controlled synthetic regime-champion streams (each regime has a distinct optimal method, with a tunable separation parameter) isolate mechanisms that real data confounds.

Baselines. Homogeneous (best single method), random selection, cooperative MARL (IQL, VDN, QMIX, MAPPO) at matched agent counts and training budgets, and—for the capacity experiments—the five-arm comparison of Section 6.

Metrics. Specialization Index (organization), task prediction error (performance), post-reactivation error (retention, measured in a 15-step probe window after each regime reactivation). Protocol: 20–30 seeded trials per condition, 95% confidence intervals, one-sided Welch tests with Holm–Bonferroni robustness checks for sweep families.

8.2 Results on Emergent Self-Organization

Under the default configuration every domain converges to near-perfect organization (mean SI = 0.992; all domains > 0.98), versus ≤ 0.02 for all cooperative MARL baselines—a qualitative, not quantitative, gap (per-domain bars in Appendix A). The critical ablation sets $\lambda = 0$: competition alone yields mean SI = 0.65 (every domain > 0.49), 5.1× the random baseline, confirming that the niche bonus accelerates but does not cause specialization. Populations use 86% of the method inventory on average. We treat SI as a *diagnostic, not a headline*: it is invariant to regime-label shuffling and inflatable by the learning rate (Section 8.8), so high SI certifies organization, not exploitable structure. The performance-relevant results follow.

8.3 Results on Coverage Under Bounded Capacity

With per-agent capacity bounded ($K < R$), the specialized population beats the capacity-matched monolith decisively (synthetic +73% at K=1; real traffic +22%), and beats the equal-capacity *random*-diversity population specifically at tight capacity (synthetic +57%, $p < 10^{-16}$; traffic +8.7%, $p < 10^{-3}$ at K=1), where random assignment leaves coverage gaps that competitive exclusion provably fills (Figure 5). Against the purpose-built arms the honest finding is parity: the learned-diversity objective matches competition for $K \geq 2$ (and exceeds it on some real domains), and the MoE router matches it on the synthetic task at every K. A method-overlap sweep maps the boundary: competition's edge over learned diversity grows monotonically with method exclusivity, from −4.8% (overlapping methods) to +29.3% (fully exclusive). *Spatial coverage is a commodity*; competition's value there is parsimony and sample-efficiency on noisy real data (it beats the router on traffic at K=1 by +5.1%), not dominance.

8.4 Results on Retention Under Non-Stationarity

Regimes cycle (a sliding window of W=3 of R=6 regimes is active per epoch), so every regime goes dormant and reactivates. This is where the five arms dissociate exactly as Principle 1 predicts (Figure 6):

The population beats the capacity-matched monolith by +33% overall and +71% in the post-reactivation window ($p < 10^{-36}$ at K=3), and the router—which matched competition on spatial coverage—fails here decisively (0.928 vs. 0.283 at K=1, $p \sim 10^{-35}$; the gap stays significant through K=4, $p < 10^{-12}$), closing only as $K \to R$. The specialized population is essentially flat in K: even K=1 agents collectively form a complete persistent memory.

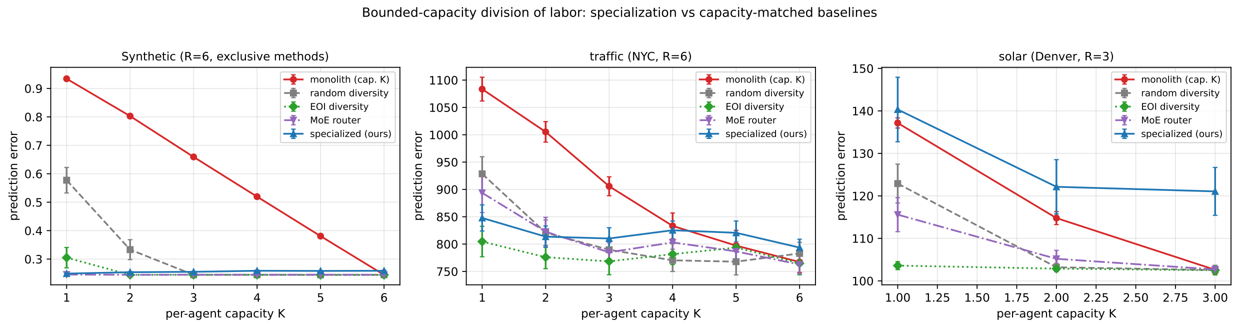

Figure 5 Coverage under bounded capacity: task error versus per-agent capacity K for the five allocation arms (95% CIs), on the synthetic regime-champion stream and real traffic data. The specialized population dominates at tight capacity; purpose-built baselines close the gap as K grows—coverage is a commodity, and competition's edge is parsimony.

Arm (K=1, hard model)	**Post-reactivation error**	**Retains?**
Monolith	1.015	no
MoE router	0.928	no
Random diversity	0.603	partially (coverage gaps)
Learned diversity	0.322	yes
GAUSE	0.283	yes
Oracle fixed (skyline)	*0.253*	*yes (given labels)*

Table 2 Post-reactivation error at K=1 under the hard (LRU) capacity model. The two reward-driven arms cluster at the top (forgetting); the reward-independent arms retain—fixed random niches only partially, because at K=1 they may leave a regime uncovered, while learned diversity and GAUSE cover and retain in full. The italicized last row is the oracle fixed-assignment *skyline*: handed both the labels and a perfect covering permutation, it needs no discovery and marks the irreducible floor (0.253, the noise level). GAUSE comes within 0.030 of this skyline at K=1 *without being given the assignment*, and is statistically indistinguishable from it by K=3 (detail in Section 8.5; oracle skyline in Figure 6).

8.5 The Honest Competitor: an Oracle Fixed Assignment

A reader who has followed the setup will object that, since every arm is handed the regime label r_t, the trivial baseline is not random diversity but an *oracle fixed assignment*: pin agent i to a covering niche by hand—at K=1 the identity permutation agent $i \mapsto$ regime i—with no learning of *where* to specialize. This arm is reward-independent and perfectly covering *by construction*, pays zero discovery cost, and so is the strongest possible member of the retaining class. The natural question is then sharp: *if I know R and the labels, why do I need emergent competition at all—why not just hand-assign?*

We answer it directly by running the arm (the black dashed skyline in Figure 6). The oracle is a *skyline*, not a competitor: it cannot be beaten on retention because it already has everything (labels, a perfect partition, no learning to do). The result is that GAUSE *matches it without the assignment*. At K=1 the oracle floor is 0.253 (the intrinsic noise level) and GAUSE reaches 0.283—a 0.030 gap that is the measurable price of *discovering* the covering permutation from scratch rather than being told it. By K=3 the gap closes to 0.006 (GAUSE 0.260 vs. oracle 0.254), statistically indistinguishable (Welch $p \approx 0.05$), and both sit at the floor while the reward-driven arms are still forgetting (0.906 and 0.719). So the honest reading is: the value of emergent competition is *not* that it beats hand-assignment given the labels—it cannot—but that it (i) recovers essentially all of the oracle's retention automatically, with no pre-commitment to a partition, and (ii) degrades gracefully off the $N \approx R$ diagonal where a hand-designed permutation is no longer available (Section 8.9). Competition buys you the oracle's assignment when you are unwilling or unable to design it yourself; the latent-regime result (Section 8.7) is what lets it do so without the label too. The GAUSE≈oracle match is also not an artifact of the eviction rule: under the soft interference model the two are indistinguishable at *every* K (differences ≤ 0.002, $p > 0.13$; at K=1, oracle 0.253 vs. GAUSE 0.254),

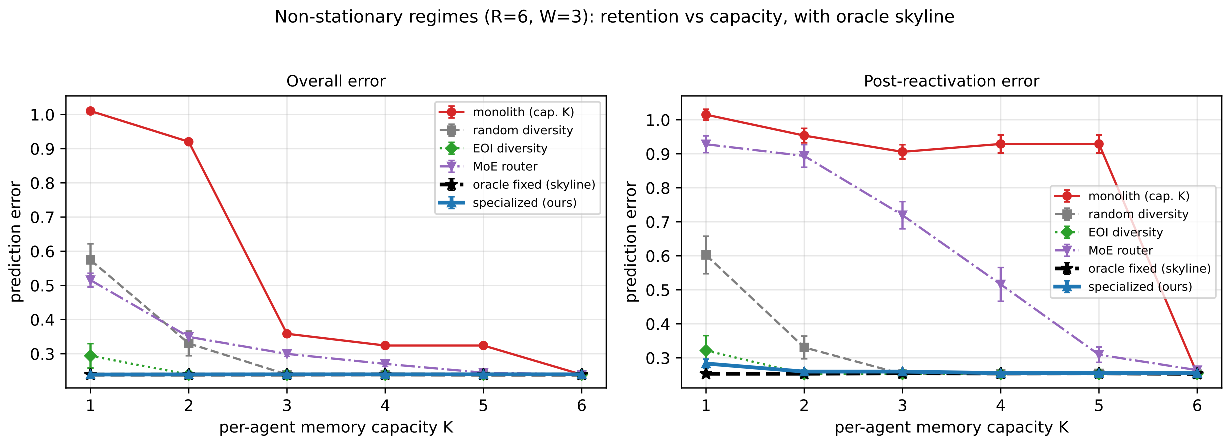

Figure 6 Retention under non-stationarity (hard LRU model, 95% CIs): overall and post-reactivation error versus capacity K for the five mechanism arms plus the oracle fixed-assignment *skyline* (black dashed). The reward-driven arms (monolith, MoE router) pay a persistent forgetting penalty that closes only as $K \to R$; the reward-independent arms are flat in K and track the oracle floor, with GAUSE (blue) recovering the privileged skyline without being given the assignment (Section 8.5).

even tighter than under hard LRU.

8.6 Ablation: Robustness to the Memory Model

To confirm the dissociation is not an artifact of discrete LRU eviction, we re-run the entire sweep under the soft interference model (no eviction; graded decay). The picture is unchanged—the monolith forgets, the population retains (≈ 0.25, a $+70\%$ gap, $p \sim 10^{-40}$), and the reward-driven router stays significantly worse than the population ($+36\%$ at $K{=}1$; $+16\%$ at $K{=}3$), degrading more gracefully but never closing the gap until $K \to R$. Full sweep and figure in Appendix B; the separation is a property of reward-driven allocation, not of any particular forgetting mechanism.

8.7 Dropping the Oracle Label: Class-Incremental GAUSE

Every result so far is task-incremental: each arm is handed the regime label r_t. The deepest question a reviewer can ask is whether the central retention claim survives when that label is removed—the genuinely hard *class-incremental* setting, where which regime is active must be inferred. We test it directly. The key realization is that the label is not actually needed: in this environment regime r is signalled by method r being best, and *which method an agent played is observable.* So we let each agent carry its niche affinity over the *method* space—a proxy for the latent regime—instead of over regimes. An agent learns "I am the M_3 specialist" without ever being told "this is regime R_3." Because winner-take-all selects on realized quality, the agent whose specialty method matches the active regime wins: the winner *is* the implicit regime estimate, and the true regime is used only to score retention, never to drive any arm.

The result (Figure 7) is that the dissociation survives label removal. Label-free GAUSE self-organizes against the *hidden* regimes (specialization index 0.75, coverage 0.86—imperfect, with occasional collisions, which is the honest signature of unsupervised recovery) and retains: its post-reactivation error is 0.38 at $K{=}1$ and flat in K, versus 1.07 for a label-free capacity-bounded monolith that forgets ($+65\%$, the same separation the labelled experiment shows). The price of dropping the label is real but modest—0.38 label-free versus 0.283 label-given—and far smaller than the gap to the forgetting baseline. Reactivation is detected purely from input similarity: when a dormant regime returns, its method works again and the intact specialist wins, with no label and no change-point detector. This converts the class-incremental extension from a promissory note into a demonstrated result, and is the strongest evidence that retention is a property of reward-independent self-organization rather than of privileged access to the regime identity.

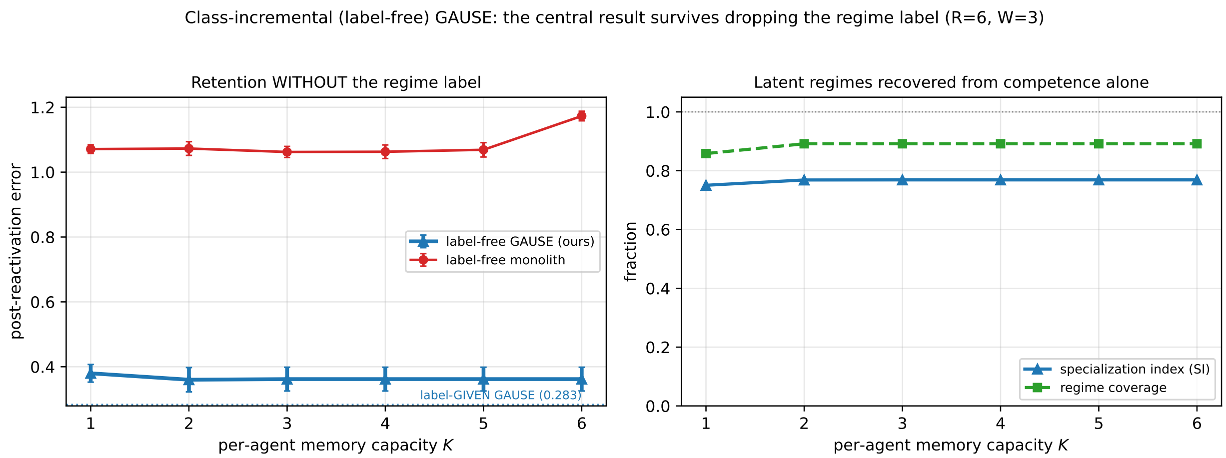

Figure 7 Class-incremental (label-free) GAUSE, with the regime label *removed* from every arm (R=6, W=3, hard LRU, 95% CIs). **Left:** post-reactivation error—label-free GAUSE (≈ 0.38, flat in K) retains while a label-free monolith forgets (≈ 1.07); the dotted line is the label-given GAUSE reference (0.283), the modest price of inference. **Right:** the population recovers the latent regimes from competence alone (specialization index 0.75 and coverage 0.86 at K=1, rising slightly with K)—imperfect but clearly structured, the honest signature of unsupervised partition discovery.

What this does and does not show (read this). The method-space proxy is exact *by construction* here: in our synthetic stream regime r is signalled by method r being uniquely best, a clean one-to-one correspondence, and the played method is observable. So this experiment removes the *label* but not the assumption that a low-dimensional, observable signature separates the regimes—it is honest class-incremental inference only to the extent that such a signature exists. The part that must generalize is precisely this: on real, feature-rich data with overlapping or non-identifiable methods the proxy is no longer exact, and recovering the latent regimes becomes a genuine online clustering problem (the open correspondence question of Section 10). We therefore read Figure 7 as a proof of concept that the competitive dynamics *can* self-supervise the partition when a separating signal is present, not as evidence that they will on arbitrary inputs.

8.8 Ablation: The Honest Audit

Three controls scope all of the above. **(i)** SI is invariant to regime-label shuffling: it measures organization, not structure. **(ii)** An oracle structure-diagnostic shows four of six domains have a globally dominant method (no exploitable regime structure); on the two structured domains (traffic, solar), causal leakage-free regime labels beat shuffled labels by +15.4% and +12.8%. **(iii)** With *unlimited* capacity, a single regime-conditioned monolith matches or beats the population: division of labor is a task-accuracy lever *specifically* under bounded capacity and non-stationarity, not in general. We consider this audit as much a contribution as the positive results: it states exactly where the mechanism's value lives.

8.9 Closing the Design Space: Function Approximation, a Hybrid Router, Drift, and Population Sizing

Four further experiments close the gaps a careful reader will press on: is the router's failure an artifact of tabular learners, can a reward-driven router be *fixed*, what happens when a regime returns *changed*, and what happens when the population size leaves the $N \approx R$ diagonal?

The router's forgetting is architecture-agnostic (Figure 8). Replacing the tabular learners with *gradient-trained neural experts* (MLPs trained by SGD) on permuted-digits—a standard continual-learning benchmark of R=5 pixel-permutation tasks, with the number of experts E as the capacity analog—reproduces the dissociation in full. At E=R the reward-driven MoE router's post-reactivation test error is 0.576 versus GAUSE's 0.218 (+62% lower, $p \sim 10^{-9}$), and the router does *not* improve with more experts (its still-

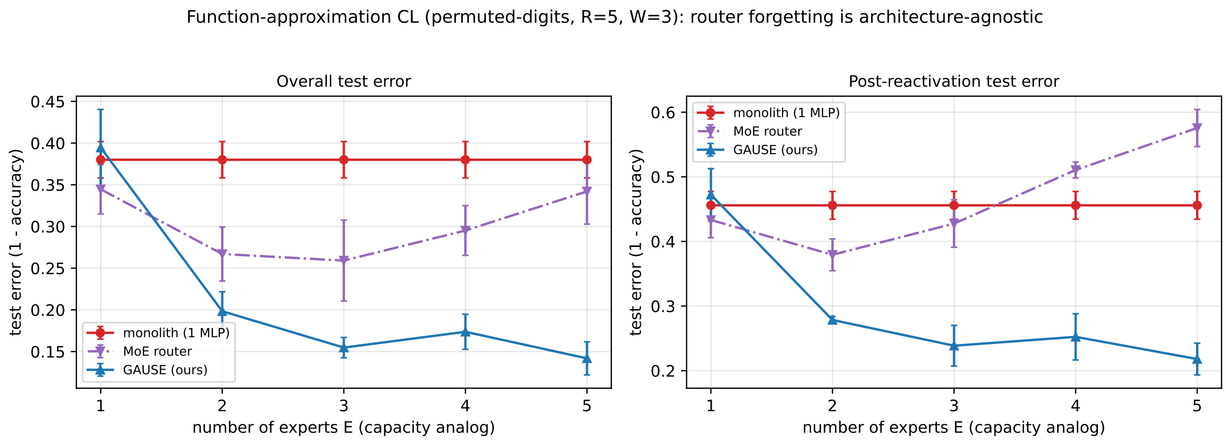

Figure 8 Function-approximation continual learning with gradient-trained MLP experts (permuted-digits, R=5). The reward-driven router (purple) forgets dormant tasks and does not improve with more experts E; GAUSE (blue) retains them (+62% lower post-reactivation error at E=R). The router's failure is architecture-agnostic.

learning gate keeps reallocating capacity to active tasks). The forgetting is thus a property of reward-driven assignment, not of the tabular substrate—the evidence the agentic-AI / MoE application narrative needs.

The dissociation survives on a standard image benchmark (Figure 9). Because the gas-sensor reality check (Section 8.10) showed the forgetting story belongs to the representation-sharing regime rather than to linear-on-features, we put exactly that surviving claim on a benchmark the continual-learning community recognizes: *Split-CIFAR-100* with small convolutional experts trained by SGD (R=5 disjoint 10-way tasks streamed through the same sliding-window recurrence schedule, so dormant tasks genuinely reactivate). The dissociation reproduces: at E=R GAUSE's post-reactivation test error is 0.581 versus the reward-driven router's 0.748 (+22% lower, $p \sim 10^{-3}$), and—the diagnostic signature—GAUSE closes the gap monotonically as capacity grows (0.793 → 0.581 from E=1 to E=R) while the router barely moves (0.807 → 0.748) and the monolith is flat (0.802). The edge is *smaller* than on permuted-digits (+22% vs +62%): on harder real images with a small CNN and a short per-visit training budget, less is retained by anyone, so there is less retention to dissociate—we report the attenuation rather than the headline. The mechanism, however, transfers intact to a standard benchmark, which is what the claim needed.

A hybrid router with a reservation term (Figure 10). Keeping the reward-driven gate but adding an explicit reservation—once the gate commits a regime to an expert, that expert pins the regime's beliefs even while dormant—recovers most of the lost retention *when there is spare capacity*: at K=3 the post-reactivation error falls from the vanilla router's 0.719 to 0.374 (+48%, $p \sim 10^{-11}$), approaching GAUSE (0.260). But the fix *cannot* help at K=1 (0.928, identical to vanilla): a single hard slot cannot both serve the active regime and reserve a dormant one, whereas the N=R competitive population retains even at K=1 (0.283) by distributing memory across idle specialists. Reward-*independence of the protection* is the operative property; competition obtains it for free, with no reservation bookkeeping and no spare-capacity requirement.

Intra-regime drift: when retention backfires (Figure 11). Finally we stress the win-only EG update directly: we let each regime's optimal method *drift* on every reactivation. Standard GAUSE pays for its stability—a converged specialist holds its niche by identity and serves it with stale beliefs, giving an overall error of 0.90, *worse* than a K=3 monolith (0.35) that is forced to relearn and stays current. This is the predicted "stale specialist" failure, now quantified. A lightweight *staleness trigger* (reset a niche whose owner is confidently underperforming—drift, not noise—and soften its affinity to re-open competition) recovers most of it (0.43 overall, $\sim$ 53% lower); the residual post-drift penalty is the irreducible relearning cost. Reward-independent retention is an asset under dormancy and a liability under drift, and a cheap trigger restores adaptivity without sacrificing the dormancy benefit.

Population sizing off the $N \approx R$ diagonal. Sweeping N on the retention protocol characterizes both off-diagonal regimes (full sweep and figure in Appendix C). The deployment-relevant summary: with *scarce*

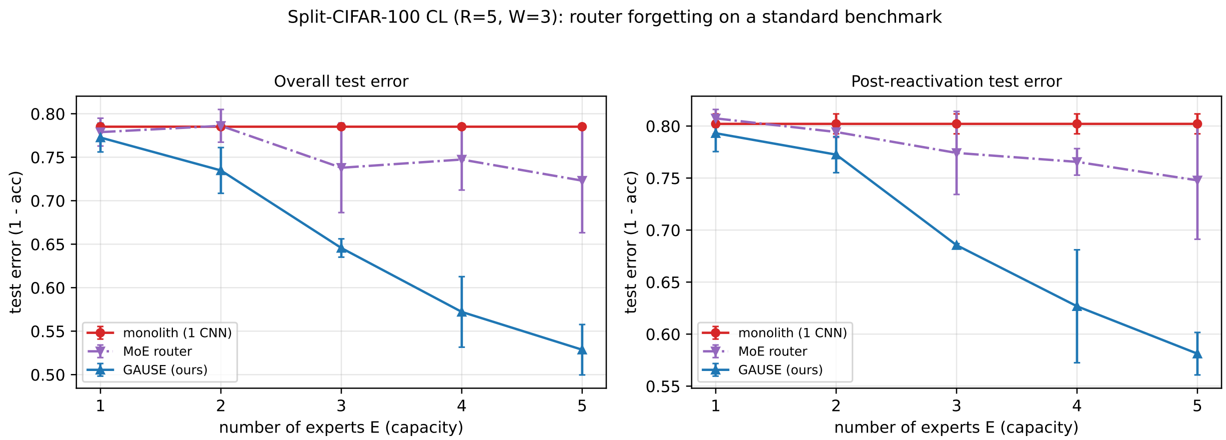

Figure 9 Split-CIFAR-100 with gradient-trained CNN experts (R=5 disjoint 10-way tasks, sliding-window recurrence, 4 trials, 95% CIs)—the surviving representation-sharing claim on a standard benchmark. GAUSE (blue) lowers both overall and post-reactivation test error monotonically as capacity E grows; the reward-driven router (purple) barely improves and the monolith (red) is flat. At $E{=}R$ GAUSE retains +22% better than the router (0.581 vs 0.748, $p \sim 10^{-3}$)—a smaller but significant edge than on the easier permuted-digits, honestly attenuated by the harder images and short training budget.

agents ($N < R$) coverage is governed by N, not by total capacity $N{\cdot}K$ (winner-take-all gives each agent one primary niche, so a larger K does *not* rescue coverage); with *abundant* agents ($N \gg R$) coverage saturates and exactly $N - R$ agents idle harmlessly. The rule is therefore *provision* $N \gtrsim R$—surplus is harmless, deficit forgets.

8.10 Real Data: The Class-Incremental Claim, Tested and Bounded (UCI Gas Sensor Array Drift)

The class-incremental subsection above flagged the one assumption its synthetic result could not shed: that a low-dimensional, observable signature separates the regimes. We now test that assumption on real data—the UCI Gas Sensor Array Drift dataset (13,910 samples, 128 real sensor features, 6 gas classes, 10 batches collected over 36 months of genuine sensor drift, with *no* method↔regime bijection). We stream the batches in time order under the same sliding-window dormancy schedule and measure post-reactivation accuracy (Figure 12). The outcome is a clean, honest split—one claim holds, one is bounded, one is refuted in this regime—and we report all three, because the split is more informative than any single headline.

(1) Full-capacity linear continual learning does *not* forget here. A plain online multinomial-logistic classifier trained by naive SGD reaches post-reactivation accuracy 0.68 and overall accuracy 0.93; experience replay is identical; EWC is *worse* (0.44), because its Fisher anchor fights the real drift it should be adapting to. On these real, near-linearly-separable sensor features there is simply nothing to catastrophically forget, so there is no linear-on-features retention win to claim. The forgetting dissociation the framework *does* own lives one level up, in the representation-sharing regime: the permuted-digits experiment (Figure 8) shows the reward-driven router forgetting (+62%) with *gradient-trained* experts, where learned representations genuinely interfere, and the same dissociation reproduces on *Split-CIFAR-100* with convolutional experts (+22% at $E{=}R$, $p \sim 10^{-3}$; Figure 9)—a standard benchmark, so the claim is not an artifact of one toy dataset. Real data thus relocate—rather than refute—the catastrophic-forgetting claim: it is a property of capacity-bounded *representation sharing*, not of arbitrary feature streams.

(2) Label-free recovery collapses under real overlap. Strip the label and let each agent infer the class from the 128-d vector: GAUSE-LF post-reactivation accuracy falls to 0.00, with specialization index 0.53 and coverage 0.83 (five of six classes ever owned). Critically, the label-free *monolith* also scores 0.00—the "GAUSE retains, monolith forgets" dissociation that the synthetic 1:1 signature produced *vanishes*. The mechanism is visible in the traces: when a dormant class returns, its drifted features are nearest to a *currently active* class's prototype, so the implicit class estimate is simply wrong. The method↔regime proxy

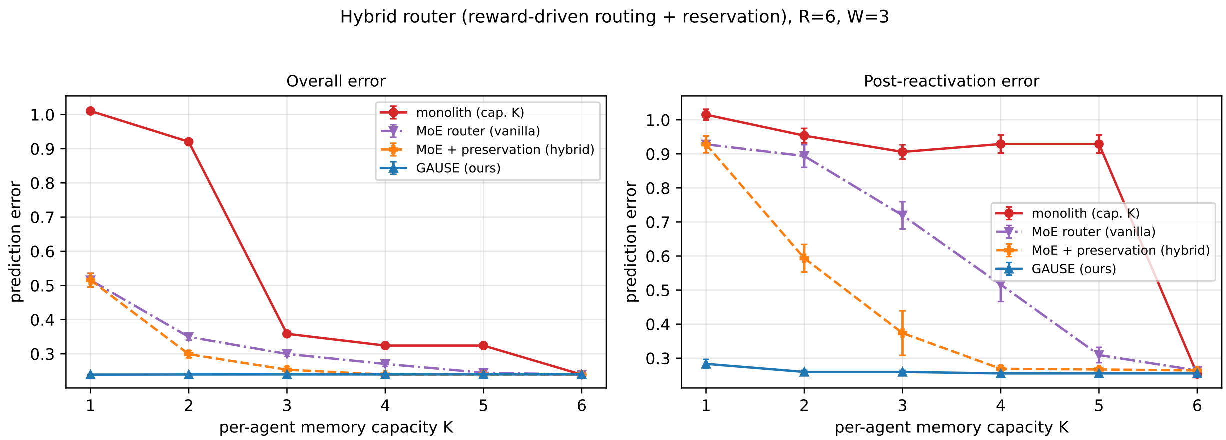

Figure 10 Hybrid arm: reward-driven routing *plus* a reservation term. It recovers most retention with spare capacity (+48% at K=3) but cannot help at K=1; the competitive population retains even at K=1. Reward-independence of protection is the operative property.

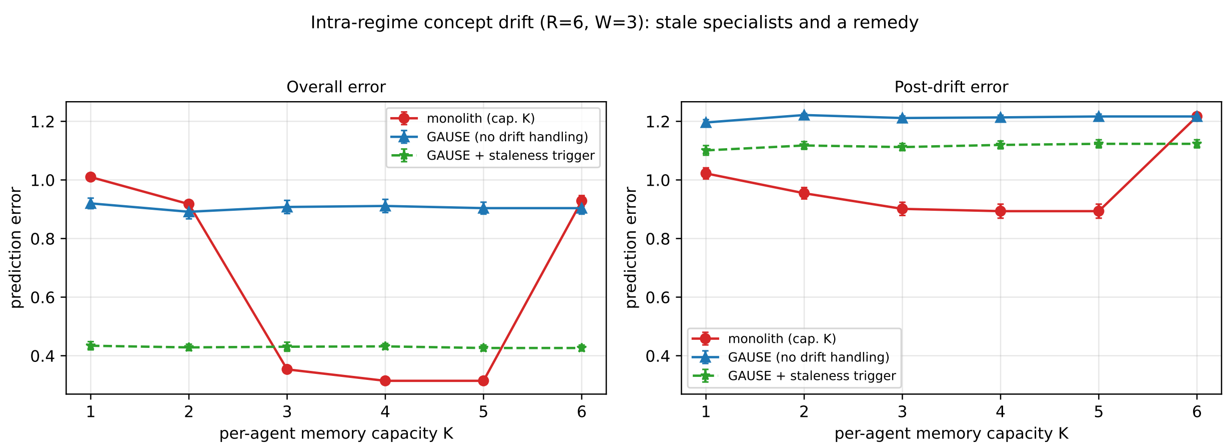

Figure 11 Intra-regime concept drift. Standard GAUSE (blue) pins a stale specialist (overall error 0.90, worse than a relearning monolith); a staleness trigger (green) recovers most of it (0.43). Reward-independent retention helps under dormancy and hurts under drift.

was load-bearing after all. This refutes the strong reading of the label-free headline on real overlapping drift and confirms—rather than weakens—the scope we stated in Section 8.7: class-incremental GAUSE is a proof of concept for *separable-signal* regimes, not for arbitrary high-dimensional inputs.

(3) Robustness, honestly inverted. On the benign synthetic surrogate the supervised baselines sit flat at 1.00 (nothing to forget), so that sweep is uninformative. On real drift the sweep *is* informative, but not in GAUSE's favor: replay is flat at 0.68 across buffer sizes; EWC degrades monotonically as its anchor tightens (0.68 at λ=0, through 0.44 at λ=20, into numerical divergence by λ=300); and GAUSE-LF is flat at 0.00. The only robustness point that survives is narrow and real—EWC's single knob can actively hurt and must be kept near zero on a drifting stream, whereas competition has no such knob—but here GAUSE's flat line sits *below* the competitors, not above.

What this does to the contribution. Taken together the real data move the paper off the performance axis. On a genuine recurring-regime stream the parsimony-*as-performance* reading does not hold: a one-line classifier with labels beats label-free GAUSE outright, and full-capacity continual learning does not forget. What survives contact with real data is the *mechanism*: (i) the reward-independence principle and its router-forgetting prediction, confirmed with gradient-trained experts where representations actually interfere; and (ii) the honest map of *where* competition helps—bounded capacity, representation interference, and separable

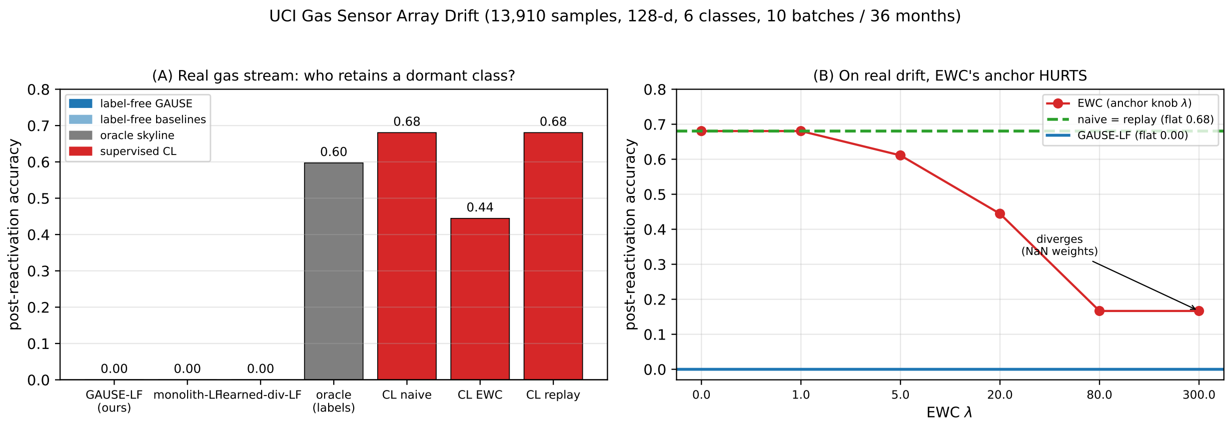

Figure 12 Real data (UCI Gas Sensor Array Drift). **(A)** Post-reactivation accuracy by arm. All label-free arms—GAUSE included—collapse to 0: a dormant class's drifted features are routed to an active class's prototype, so the dissociation against the label-free monolith *disappears*. A full-capacity supervised linear classifier (naive/replay) does not forget (0.68); EWC's anchor hurts under drift (0.44). **(B)** The EWC λ sweep on real drift: anchoring monotonically degrades retention and eventually diverges—the opposite of its flat 1.00 on the benign surrogate. The honest reading: on real near-linearly-separable features there is nothing to forget, so the catastrophic-forgetting claim belongs to the representation-sharing neural regime (Figure 8), and the label-free claim is bounded to separable-signal regimes.

regime signals—and where it does not. We therefore read GAUSE as a mechanism/lens contribution rather than a benchmark winner, and have aligned the abstract and the limitations accordingly.

9 Potential Applications

GAUSE is a mechanism, not a model: it applies wherever (i) the environment decomposes into recurring regimes, (ii) per-learner capacity is bounded, and (iii) regimes go dormant and return. Those three conditions are remarkably common. This section surveys the application domains where we believe the mechanism—and especially its retention guarantee—has the most leverage. Table 3 summarizes the mapping; a deployment recipe follows at the end of the section.

Evidence status (read this first). The discussion below is *prospective*, and we are explicit about how far our evidence reaches. Two domains—*quantitative finance* (crypto, commodities) and *distributed sensing* (solar, weather, air quality)—use data drawn from the same families as our validation domains, so the self-organization and bounded-capacity results transfer most directly there. The agentic-AI / MoE memory case sits one tier below: our function-approximation experiment (Section 8.9) shows the router's forgetting and GAUSE's retention with *gradient-trained neural experts* on a standard continual-learning benchmark, so the mechanism-level prediction is no longer an analogy—but we have not deployed it inside a production-scale MoE or agent framework, so the systems claim remains untested. The remaining domains (robot fleets, cloud serving, clinical monitoring) are *conjectures*: they satisfy the three structural conditions on paper, but we have run no experiments in them. We present the full survey because the mechanism's preconditions are domain-agnostic, but the reader should treat everything past the first two subsections as a research agenda, with the MoE case the only one backed by direct mechanism-level evidence.

9.1 Quantitative Finance: Regime-Switching Strategy Pools

Financial markets are the canonical regime-switching environment—trending, mean-reverting, high-volatility, and crisis regimes recur on irregular schedules [10]—and two of our six validation domains (crypto, commodities) are financial. The standard industry practice is reward-chasing by construction: capital and research attention flow to the strategies that performed recently, and strategies that lose relevance are retired. Our results identify the precise cost of that practice. A crisis regime may be dormant for years; a

Field	**Regimes**	**Learners**	**Dormancy risk (what is forgotten)**
Agentic AI memory	task/skill families	skill modules, expert pools	rarely-invoked skills overwritten by frequent ones
Quantitative finance	market regimes (trend, crisis, range)	strategy pool / capital sleeves	crisis playbooks decay during calm markets
Edge sensing & IoT	seasonal/diurnal patterns, fault modes	on-device models	rare fault signatures lost between occurrences
Robot fleets	zones, task types, failure modes	robots / controllers	seldom-used manipulation skills degrade
Cloud serving	workload classes	autoscaled worker pools	warm capacity for bursty rare workloads
Clinical monitoring	patient subpopulations, rare conditions	model ensemble	rare-condition detectors drowned by common cases

Table 3 Mapping GAUSE onto application domains. In every row the dormancy column is the failure a reward-driven (usage-driven, profit-driven, traffic-driven) allocator suffers and a reward-independent population structurally avoids.

capital allocator driven by recent P&L (the financial analogue of a learned gate) starves the crisis specialists, their infrastructure and calibration decay, and when the regime reactivates the desk pays the full relearning cost—at the worst possible time. A GAUSE-style strategy pool inverts this: strategies compete for regimes, the winner-take-all dynamics assign each strategy a niche, and converged specialists are *retained through dormancy by identity*, not by P&L. The niche bonus λ has a natural reading as a diversification mandate, and the capacity bound K models the real constraint that a single desk cannot stay calibrated to every market state. Three concrete uses: (i) a **regime-specialist ensemble** for return forecasting, where the dormant-regime specialist (e.g., the short-volatility-crash model) is kept warm; (ii) **market-making parameter banks**, where quoting profiles specialized to liquidity regimes are retained across regime cycles; and (iii) **risk-model committees**, where stress-period models must survive long calm stretches—exactly the retention property reward-driven model selection lacks.

9.2 Distributed Sensing and Edge Intelligence

Sensor networks and edge deployments are populations of capacity-limited learners by physical necessity: each device has tight memory and compute, and the environment it models has strong seasonal and episodic structure (diurnal cycles, weather patterns, rare fault signatures). Centralized coordination is expensive precisely where edge deployment is attractive. GAUSE needs only a local competition signal (which device best predicted the current window), so a fleet can partition pattern-space without a coordinator: some devices become storm specialists, others baseline-drift specialists. The retention property matters most for **rare-event detection**: a fault signature seen once a year is a dormant regime, and a usage-driven model update schedule will overwrite it. Our solar, weather, and air-quality domains are direct evidence that the mechanism organizes correctly on exactly this kind of data.

9.3 Other Candidate Domains

The remaining domains in Table 3 share the same three preconditions but we have run no experiments in them; we keep them brief and explicitly conjectural. **Agentic-AI / MoE memory** is the case our theory engages most directly: agent skill libraries and MoE expert pools both face bounded capacity and a long tail of rarely-invoked competences, and any recency- or utility-driven eviction (LRU caches, usage-weighted distillation, gates trained on task loss) is a reward-chasing allocator that Principle 1 predicts will overwrite the skill needed *next quarter* because it emits no usage signal *today*; letting experts compete and then pinning the converged assignment realizes the gate-freezing prescription of MoE continual-learning theory [15] without a freeze schedule. This case is backed by our function-approximation continual-learning experiments on two standard benchmarks (permuted-digits and Split-CIFAR-100; Section 8.9), the evidence here that goes beyond analogy. **Multi-robot fleets** (zones/task-types as regimes; battery, gripper, and local-map budgets as capacity) could partition a floor through win-driven affinity with no message passing, keeping seldom-used skills warm. **Cloud serving** maps cleanly onto our arms—autoscaling is the monolith, a learned request router is the MoE arm, static over-provisioning is random diversity—and a competitive worker pool would pin warm specialists for bursty rare workload classes against the cold-start penalty. **Clinical monitoring** treats patient subpopulations or rare conditions as regimes, dedicating ensemble members to rare presentations that aggregate-performance updates would otherwise drown out. In all four, the structure diagnostic (Section 8.8) is the prerequisite test: if one model dominates everywhere, the population adds nothing.

9.4 When Not to Use It

The audit section is a user manual in miniature, and we restate it as deployment guidance. GAUSE does *not* pay when: (i) capacity is effectively unlimited—a single regime-conditioned model then matches the population with less machinery; (ii) the environment has no regime structure (one method dominates globally)—our oracle diagnostic detects this case in advance; or (iii) regimes never go dormant—without non-stationarity, retention is free and reward-driven routing is a fine choice. The mechanism's value concentrates precisely where all three conditions bind, which the applications above are selected to satisfy.

Two failure modes the mechanism does not yet handle. Beyond those scope limits, the converged assignment has two known blind spots that a deployer must weigh. *(a) Within-niche drift.* The EG affinity update moves identity only on *wins* (positive evidence of fit), so a specialist does not spontaneously relinquish a niche it is silently getting worse at: if a regime's optimal method *drifts* while the specialist idles, the converged assignment pins a stale specialist in place. We now test this (Section 8.9, Figure 11): under drift, standard GAUSE's overall error (0.90) is *worse* than a relearning monolith's (0.35), and a lightweight staleness trigger that re-opens competition recovers most of it (0.43). The remedy works but leaves a residual relearning cost, and we have not characterized drift rate or detection threshold in general—so drift remains a deployment risk to weigh. *(b) Off-diagonal population sizing.* Our clean one-per-regime equilibrium and coverage guarantee are *proved* for $N \approx R$; the off-diagonal behavior we now *measure* (Section 8.9, Figure 15). Scarce agents ($N < R$) leave regimes uncovered—and crucially, coverage tracks N rather than total capacity $N \cdot K$, so a bigger per-agent budget does not help—while abundant agents ($N \gg R$) idle harmlessly. The remaining gap is theoretical (no proof of the off-diagonal equilibrium) and mechanistic (letting scarce agents fill $K > 1$ slots).

> **Deployment recipe.** (1) Define regimes: a discrete, recurring partition of the input stream (volatility bands, workload classes, seasonal modes). (2) Define the quality signal: any per-step scalar comparable across learners. (3) Set capacity K honestly: how many regimes can one learner stay calibrated on? (4) Run winner-take-all competition with the EG affinity update ($\eta \approx 0.6$, $\lambda \in [0.2, 0.4]$ or 0 for the pure mechanism). (5) Read out the converged assignment and *pin it*: the assignment, not the gate, decides who holds what. (6) Re-open competition only when the regime inventory itself changes.

Honest accounting of what "no machinery" means. We claim competition needs no gate, no freezing schedule, and no diversity coefficient, and that is true—but the recipe makes clear that effort is *relocated*, not eliminated. Steps (1)–(3) still demand the practitioner supply a discrete regime partition, a cross-learner-comparable quality signal, and a capacity budget K; these are real design decisions, and on a novel domain the regime partition can be as hard to get right as a gate. Step (6) is, in particular, a *freeze trigger*: pinning the assignment until the regime inventory changes is exactly a freeze, except that the condition is "inventory changed" rather than "step count exceeded." Competition's genuine advantage is that the freeze happens *automatically and at the right time* (when competition converges) rather than on a schedule the practitioner must tune—not that the notion of freezing has disappeared.

10 Conclusion and Future Work

We presented GAUSE, a minimal framework in which winner-take-all competition over a niche-affinity simplex makes a population of capacity-bounded learners self-organize into one-per-regime specialists—with no communication, shared rewards, gates, or diversity objectives. The framework's deepest property is that its converged assignment is *reward-independent*: specialists hold niches by identity, idle through dormancy, and thereby retain what reward-driven allocators—including the standard learned MoE router—systematically forget. Across five capacity-allocation mechanisms, retention tracks reward-independence of assignment five-for-five, and an oracle fixed-assignment skyline—handed both the labels and a perfect covering permutation—confirms the ceiling: GAUSE recovers it to within 0.030 at $K=1$ and indistinguishably for $K \geq 3$ without being given the assignment. Across six real-world domains and controlled synthetic tasks, competition matches purpose-built diversity and routing baselines on spatial coverage with none of their machinery, and separates from them decisively on retention (0.283 vs. the router's 0.928 at $K=1$, $\sim$70% lower, $p \sim 10^{-35}$, robust to the memory model). An honest audit bounds the claim: with slack capacity and stationary regimes, a regime-conditioned monolith suffices; division of labor pays specifically when capacity binds and the world shifts.

Section 8.9 closes four of the gaps this document is explicit about. **(i)** A function-approximation demonstration confirms the router's failure is architecture-agnostic on *two* standard benchmarks: with gradient-trained MLP experts on permuted-digits (+62%) and convolutional experts on Split-CIFAR-100 (+22%, $p \sim 10^{-3}$), the reward-driven router forgets while GAUSE retains at $E=R$, converting the central prediction—and the agentic-AI/MoE application story—from analogy into evidence. **(ii)** A hybrid allocator (a reward-driven router *plus* a reservation term) recovers most of the lost retention when spare capacity exists but not at $K=1$, confirming that reward-*independence of protection* is the operative property and that competition obtains it for free. **(iii)** An intra-regime drift experiment quantifies the stale-specialist risk (standard GAUSE worse than a relearning monolith under drift) and shows a lightweight staleness trigger recovers most of it. **(iv)** A population-sizing sweep characterizes both sides of the $N \approx R$ diagonal: surplus agents idle harmlessly, while scarce agents under-cover in proportion to N (not $N \cdot K$), yielding the deployment rule *provision* $N \gtrsim R$.

A real-data test on the UCI Gas Sensor Array Drift stream (Section 8.10) bounds the headline honestly: on near-linearly-separable real features a full-capacity online classifier does *not* catastrophically forget, so the retention dissociation is a property of the representation-sharing regime—confirmed by the two neural benchmarks above—rather than of arbitrary feature streams, and the label-free recovery collapses under real overlap. On real data, accordingly, the contribution is a *mechanism and lens*, not a performance win—which we regard as the more durable claim.

Several directions remain. **First**, drift handling beyond our trigger: characterizing drift rate, detection threshold, and the dormancy–drift interaction, since a residual relearning cost remains. **Second**, the *theory* of off-diagonal population sizing: we now measure that scarce agents ($N < R$) under-cover (coverage tracks N, not $N \cdot K$) and abundant agents ($N \gg R$) idle harmlessly (Section 8.9), but a proof of the off-diagonal equilibrium, a mechanism letting scarce agents exploit $K > 1$, and scaling decentralized competition over hundreds of regimes with churn and partial observability all remain open. **Third**, removing the oracle regime label—which we now *demonstrate* rather than defer (Section 8.7). Letting each agent carry its niche affinity over the observable method space (a proxy for the latent regime) makes GAUSE class-incremental: it self-organizes against the hidden regimes (SI 0.75, coverage 0.86) and retains (0.38 post-reactivation at $K=1$ versus 1.07 for a label-free forgetting monolith), detecting reactivation purely from input similarity. Tested on *real*, feature-rich data with genuine overlap (UCI gas sensor, Section 8.10), however, this label-free recovery *collapses* (post-reactivation $\rightarrow 0$): the synthetic method$\leftrightarrow$regime signature was load-bearing, and the class-incremental result is bounded to *separable-signal* regimes. What remains open is therefore the genuinely hard case the gas data expose—recovering latent regimes under *non-separable* real overlap—for which the correspondence between emergent win-clusters and a stable latent labeling (online competitive-clustering / EM) is the theoretical core. **Finally**, deployment in the application domains of Section 9—trading-strategy pools, sensor networks, and expert pools inside large models—where the parsimony of competition (an automatic, correctly-timed freeze rather than a tuned schedule, and no diversity coefficient) is most valuable.

A Per-Domain Self-Organization

Figure 13 reports the Specialization Index across all six real-world domains (the summary cited in Section 8). We treat this as a diagnostic, not a headline: SI certifies organization, not exploitable structure (Section 8.8).

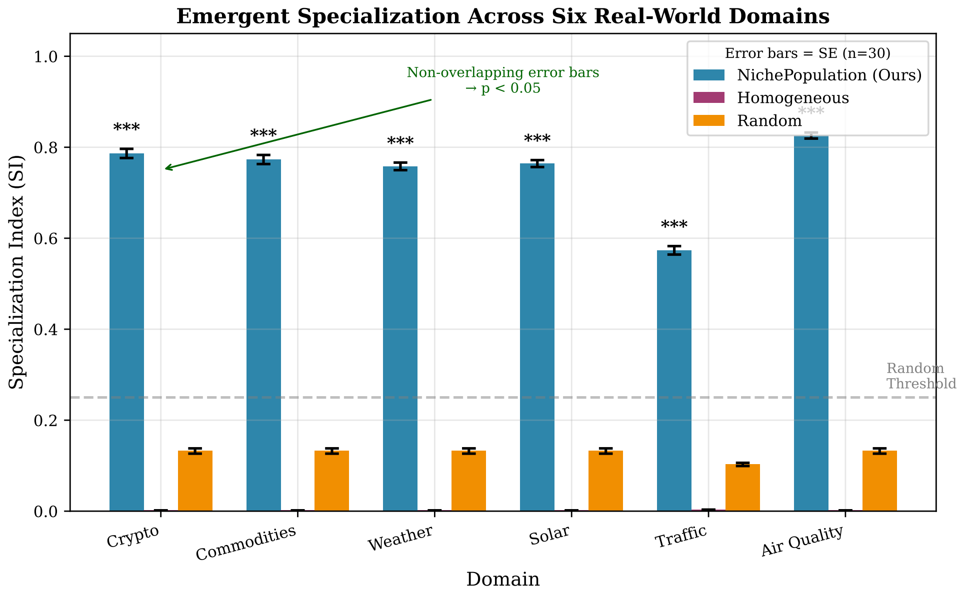

Figure 13 Emergent self-organization across the six real-world domains: GAUSE reaches SI ≈ 0.99 everywhere, while homogeneous and random baselines stay near zero.

B Robustness: Soft (Interference) Memory Model

Re-running the entire non-stationary sweep under the soft interference model (no eviction at all; every learning update on the active regime decays the pseudo-counts of the agent's other held regimes) reproduces the dissociation (Figure 14): the monolith still forgets (post-reactivation error 0.85–0.92 at $K \leq 3$), the specialized population still retains (≈ 0.25, a +70% gap, $p \sim 10^{-40}$), and the router remains significantly worse than the population (+36% at $K{=}1$, $p \sim 10^{-8}$; +16% at $K{=}3$). The router degrades more gracefully under graded interference than under hard eviction but never closes the gap until $K \to R$: the separation is a property of reward-driven allocation, not of any particular forgetting mechanism.

C Population Sizing Off the $N \approx R$ Diagonal

Our equilibrium and coverage arguments assume $N \approx R$; sweeping N on the retention protocol ($R{=}6$) characterizes both off-diagonal regimes (Figure 15). With *scarce* agents ($N < R$), coverage is capped and governed by N, not by total capacity $N \cdot K$: winner-take-all gives each agent one primary niche, so at $N{=}3$ coverage is only $\approx$ 0.4–0.5 and post-reactivation error stays high *even at* $K{=}2$, where total capacity would in principle suffice—the spare slot goes unused. With *abundant* agents ($N \gg R$) coverage saturates, exactly $N - R$ agents idle, and retention does not regress. The deployment rule is therefore: provision $N \gtrsim R$ (surplus is harmless, deficit forgets), and do not expect a larger K to substitute for more agents when agents are scarce.

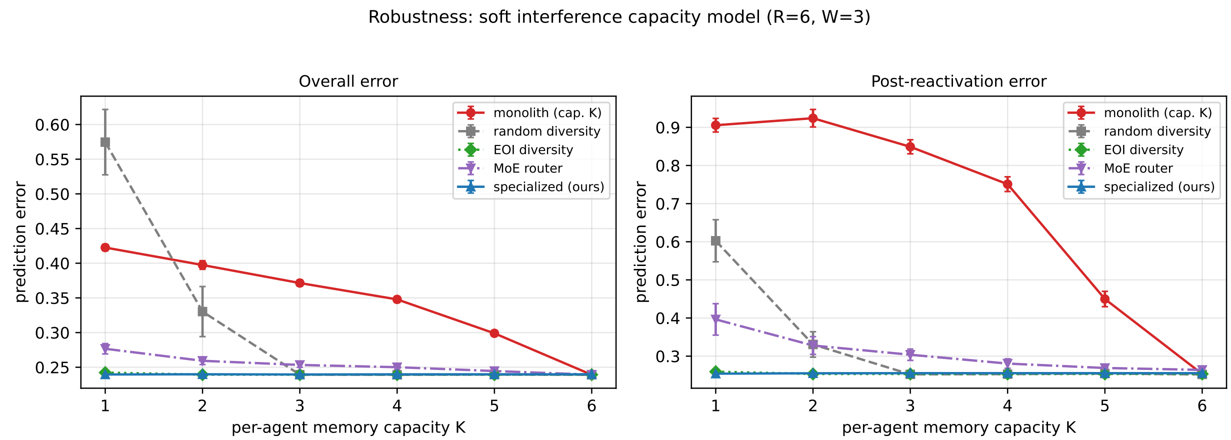

Figure 14 Robustness to the capacity model: the full non-stationary sweep re-run under the soft interference-decay model (no eviction). The reward-independence dissociation survives intact, ruling out an artifact of discrete LRU eviction.

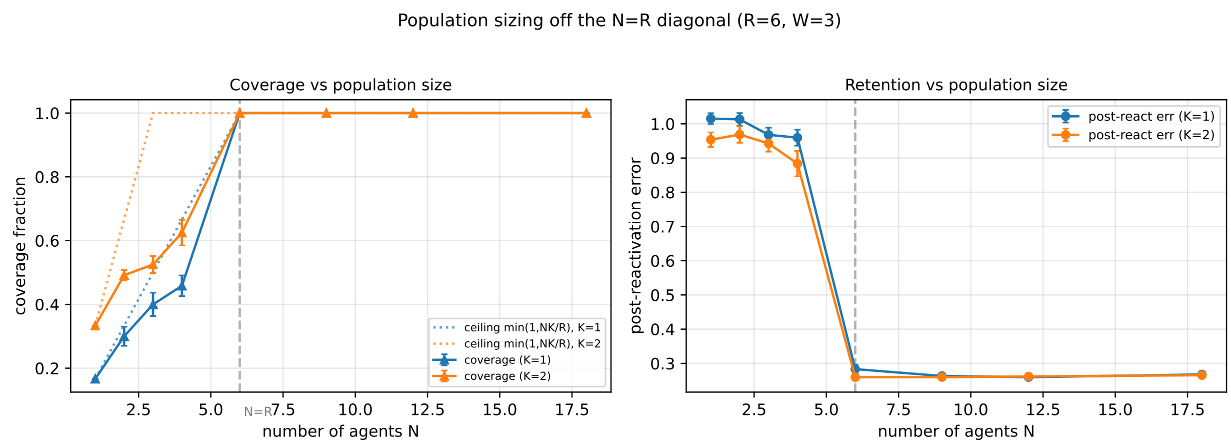

Figure 15 Population sizing off the $N \approx R$ diagonal (R=6). Below N=R, coverage (left) tracks N rather than the feasibility ceiling $\min(1, NK/R)$ and retention (right) degrades; for $N \geq R$ coverage saturates and surplus agents idle ($N - R$ redundant) without harming retention.

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